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Open rotor aircraft propulsion system with bypass flowpath

Abstract

A propulsion system for an aircraft includes an open propulsor rotor and a turbine engine. The turbine engine includes a fan section, an engine core, a first geartrain, a core flowpath and a bypass flowpath. The fan section includes a fan rotor. The engine core includes a first rotating assembly, a low pressure compressor section, a high pressure compressor section, a combustor section, a high pressure turbine section and a low pressure turbine section. The first rotating assembly drives rotation of the open propulsor rotor and the fan rotor through the first geartrain. The core flowpath extends through the low pressure compressor section, the high pressure compressor section, the combustor section, the high pressure turbine section and the low pressure turbine section. The bypass flowpath extends outside of and along the engine core.

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Background/Summary

BACKGROUND OF THE DISCLOSURE

1. Technical Field

(1) This disclosure relates generally to an aircraft and, more particularly, to a propulsion system for the aircraft.

2. Background Information

(2) Various types and configurations of aircraft propulsion systems are known in the art including those with one or more open propulsor rotors. While these known aircraft propulsion systems have various benefits, there is still room in the art for improvement.

SUMMARY OF THE DISCLOSURE

(3) According to an aspect of the present disclosure, a propulsion system is provided for an aircraft. This propulsion system includes an open propulsor rotor and a turbine engine. The turbine engine includes a fan section, an engine core, a first geartrain, a core flowpath and a bypass flowpath. The

fan section includes a fan rotor. The engine core includes a first rotating assembly, a low pressure compressor section, a high pressure compressor section, a combustor section, a high pressure turbine section and a low pressure turbine section. The first rotating assembly is rotatable about an axis and configured to drive rotation of the open propulsor rotor and the fan rotor through the first geartrain. The first geartrain is located axially between the engine core and the open propulsor rotor. The core flowpath extends through the low pressure compressor section, the high pressure compressor section, the combustor section, the high pressure turbine section and the low pressure turbine section from an inlet into the core flowpath to an exhaust from the core flowpath. The inlet into the core flowpath is next to and downstream of the fan section. The bypass flowpath extends outside of and along the engine core from an inlet into the bypass flowpath to an exhaust from the bypass flowpath. The inlet into the bypass flowpath is next to and downstream of the fan section.
(4) According to another aspect of the present disclosure, another propulsion system is provided for an aircraft. This propulsion system includes a first gear system, a second gear system, an intermediate shaft, a fan rotor, an open propulsor rotor and a rotating assembly of a turbine engine. The first gear system includes a first sun gear, a first ring gear, a plurality of first intermediate gears and a first carrier. Each of the first intermediate gears is meshed with and radially between the first sun gear and the first ring gear. Each of the first intermediate gears is rotatably mounted to the first carrier. The second gear system includes a second sun gear, a second ring gear, a plurality of second intermediate gears and a second carrier. Each of the second intermediate gears is meshed with and radially between the second sun gear and the second ring gear. Each of the second intermediate gears is rotatably mounted to the second carrier. The intermediate shaft is coupled to the first ring gear and the second sun gear. The fan rotor is coupled to the intermediate shaft. The open propulsor rotor is coupled to the second carrier. The rotating assembly is coupled to the first sun gear. The rotating assembly is configured to drive rotation of the fan rotor through the first gear system and is independent of the second gear system. The rotating assembly is configured to drive rotation of the open propulsor rotor through the first gear system and the second gear system.
(5) According to still another aspect of the present disclosure, another propulsion system is provided for an aircraft. This propulsion system includes a plurality of gear systems, an intermediate shaft, a fan rotor, an open propulsor rotor and a rotating assembly of a turbine engine. Each of the gear systems includes a planetary gear system or a star gear system. The intermediate shaft operatively couples a first of the gear systems to a second of the gear systems. The fan rotor is coupled to the intermediate shaft. The open propulsor rotor is operatively coupled to the second of the gear systems. The rotating assembly is operatively coupled to the first of the gear systems. The rotating assembly is configured to drive rotation of the fan rotor through the first of the gear systems and is independent of the second of the gear systems. The rotating assembly is configured to drive rotation of the open propulsor rotor through the first of the gear systems and the second of the gear systems.
(6) The propulsion system may also include an engine core which includes the rotating assembly. The first gear system may be arranged between the engine core and the fan rotor.
(7) The low pressure compressor section may be located axially between the first geartrain and the high pressure compressor section along the axis.
(8) The first rotating assembly may be a low pressure turbine rotor in the low pressure turbine section.
(9) The engine core may also include an intermediate pressure turbine section between the high pressure turbine section and the low pressure turbine section along the core flowpath.
(10) The low pressure turbine section may include a low pressure turbine rotor. The first geartrain may be located axially between the low pressure turbine rotor and the fan rotor.
(11) The low pressure turbine rotor may be configured to rotate independent of the first rotating assembly.
(12) The propulsion system may also include a second geartrain. The first rotating assembly may

be configured to drive the rotation of the fan rotor through the first geartrain and independent of the second geartrain. The first rotating assembly may be configured to drive the rotation of the open propulsor rotor through the first geartrain and the second geartrain.
(13) One of the first geartrain and the second geartrain may be configured as or otherwise include a planetary gear system. The other one of the first geartrain and the second geartrain may be configured as or otherwise include a star gear system.
(14) The first geartrain may be configured as or otherwise include a planetary gear system. The second geartrain may be configured as or otherwise include a planetary gear system.
(15) The first geartrain may be configured as or otherwise include a star gear system. The second geartrain may be configured as or otherwise include a star gear system.
(16) The second geartrain may be located axially between the fan rotor and the open propulsor rotor.
(17) The fan rotor may be located axially between the first geartrain and the second geartrain.
(18) At least a portion of the bypass flowpath may be annular.
(19) The propulsion system may also include a heat exchanger disposed in the bypass flowpath.
(20) The turbine engine may also include an exhaust flowpath fluidly coupling the core flowpath and the bypass flowpath to an exhaust from the turbine engine.
(21) The propulsion system may also include an open guide vane structure downstream of the open propulsor rotor. The open guide vane structure may be configured to condition air propelled by the open propulsor rotor.
(22) The propulsion system may also include a second open propulsor rotor. The first rotating assembly may also be configured to drive rotation of the second open propulsor rotor through the first geartrain.
(23) The present disclosure may include any one or more of the individual features disclosed above and/or below alone or in any combination thereof.
(24) The foregoing features and the operation of the invention will become more apparent in light of the following description and the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a sectional schematic illustration of a propulsion system for an aircraft.
(2) FIG. **2** is a cross-sectional illustration of an open propulsor blade.
(3) FIG. **3** is a cross-sectional illustration of an open guide vane.
(4) FIG. **4** is a partial schematic illustration of the propulsion system at a set of geartrains.
(5) FIG. **5** is a schematic illustration of a first of the geartrains coupled with components of the propulsion system.
(6) FIG. **6** is a schematic illustration of a second of the geartrains coupled with components of the propulsion system.
(7) FIGS. **7** and **8** are partial schematic illustrations of the propulsion system with other arrangements of the geartrains.
(8) FIG. **9** is a partial schematic illustration of the propulsion system with a single rotor propulsion module.
(9) FIG. **10** is a partial schematic illustration of the propulsion system with a multi-rotor propulsion module.
DETAILED DESCRIPTION
(10) FIG. **1** illustrates a propulsion system **20** for an aircraft. The aircraft may be an airplane, a drone (e.g., an unmanned aerial vehicle (UAV)) or any other manned or unmanned aerial vehicle or system. The aircraft propulsion system **20** is configured as an open rotor propulsion system. The

aircraft propulsion system **20** of FIG. **1**, for example, extends axially along an axis **22** between an upstream, forward end **24** of the aircraft propulsion system **20** and a downstream, aft end **26** of the aircraft propulsion system **20**. The axis **22** may be a centerline axis of the aircraft propulsion system **20** and/or one or more of its members. The axis **22** may also or alternatively be a rotational axis of one or more members of the aircraft propulsion system **20**. The aircraft propulsion system **20** of FIG. **1** includes a propulsion module **28**, a turbine engine **30** and a heat exchange system **32**.

(11) The propulsion module **28** of FIG. **1** includes an open propulsor rotor **34** and an open guide vane structure **36**. Here, the term “open” may describe a propulsion system section and/or a propulsion system component which is open to an environment **38** (e.g., an ambient environment) external to the aircraft propulsion system **20** and its propulsion module **28** and, more generally, external to the aircraft. The propulsion module members **34** and **36** of FIG. **1**, for example, are un-ducted and unshrouded components of the aircraft propulsion system **20** and its propulsion module **28**. The propulsion module **28** of FIG. **1** also includes a nose cone **40** disposed at (e.g., on, adjacent or proximate) the propulsion system forward end **24**. This nose cone **40** may be configured as a spinner which is rotatable with the propulsor rotor **34** about the axis **22**. Alternatively, the nose cone **40** may be configured as a stationary structure of the propulsion module **28**.

(12) The propulsor rotor **34** includes a rotor base **42** (e.g., a disk or a hub) and a plurality of open propulsor blades **44** (e.g., airfoils). The propulsor blades **44** are arranged circumferentially around the rotor base **42** and the axis **22** in an array; e.g., a circular array. Each of the propulsor blades **44** is connected to (e.g., formed integral with or otherwise attached to) the rotor base **42**.

(13) Each propulsor blade **44** projects spanwise along a span line of the respective propulsor blade **44** (e.g., radially relative to the axis **22**) out from an exterior surface **46** of the rotor base **42**, into the external environment **38**, to an unshrouded, distal tip **48** of the respective propulsor blade **44**. Each propulsor blade **44** is thereby configured as an un-ducted and unshrouded propulsor blade which is exposed to (e.g., disposed in) the surrounding external environment **38**. Referring to FIG. **2**, each propulsor blade **44** extends longitudinally along a mean line **50** (e.g., a camber line) of the respective propulsor blade **44** from a leading edge **52** of the respective propulsor blade **44** to a trailing edge **54** of the respective propulsor blade **44**. Each propulsor blade **44** extends laterally between and to opposing side exterior surfaces **56**A and **56**B (generally referred to as “**56**”) of the respective propulsor blade **44**. The first blade exterior surface **56**A may be a concave, pressure side surface of the respective propulsor blade **44**. The second blade exterior surface **56**B may be a convex, suction side surface of the respective propulsor blade **44**. Each of these blade exterior surfaces **56** extends longitudinally along the blade mean line **50** between (and meets at) the respective blade leading edge **52** and the respective blade trailing edge **54**. Referring to FIG. **1**, each blade element **52**, **54**, **56**A, **56**B extends spanwise from the base exterior surface **46** to the respective blade tip **48**.

(14) Each propulsor blade **44** may be configured as a fixed propulsor blade. Each propulsor blade **44**, for example, may be fixedly connected to the rotor base **42**. Alternatively, some or all of the propulsor blades **44** may each be configured as a variable propulsor blade. Each propulsor blade **44**, for example, may be pivotally coupled to the rotor base **42** so as to be operable to change, for example, a pitch of the respective propulsor blade **44**.

(15) The propulsor rotor **34** of FIG. **1** is arranged axially along the axis **22** between the nose cone **40** and the guide vane structure **36**. The propulsor rotor **34** of FIG. **1**, in particular, is arranged axially downstream, aft of the nose cone **40** and axially upstream, forward of the turbine engine **30**. An upstream, forward end of the propulsor rotor **34** may be disposed axially next to (e.g., adjacent, slightly spaced from) a downstream, aft end of the nose cone **40**. A downstream, aft end of the propulsor rotor **34** may be disposed axially next to an upstream, forward end **58** of the turbine engine **30**.

(16) The guide vane structure **36** includes an inner platform **60** and a plurality of open guide vanes **62**; e.g., airfoils. The inner platform **60** may be configured as part of (or connected to) a housing **64**

of the turbine engine **30**, or another external stationary structure of the aircraft propulsion system **20**. The guide vanes **62** are arranged circumferentially around the inner platform **60** and the axis **22** in an array; e.g., a circular array. Each of the guide vanes **62** is connected to the inner platform **60**.

(17) Each guide vane **62** projects spanwise along a span line of the respective guide vane **62** (e.g., radially relative to the axis **22**) out from an exterior surface **66** of the inner platform **60**, into the external environment **38**, to an unshrouded, distal tip **68** of the respective guide vane **62**. Each guide vane **62** is thereby configured as an un-ducted and unshrouded guide vane which is exposed to (e.g., disposed in) the surrounding external environment **38**. Referring to FIG. **3**, each guide vane **62** extends longitudinally along a mean line **70** (e.g., a camber line) of the respective guide vane **62** from a leading edge **72** of the respective guide vane **62** to a trailing edge **74** of the respective guide vane **62**. Each guide vane **62** extends laterally between and to opposing side exterior surfaces **76**A and **76**B (generally referred to as "**76**") of the respective guide vane **62**. The first vane exterior surface **76**A may be a convex, suction side surface of the respective guide vane **62**. The second vane exterior surface **76**B may be a concave, pressure side surface of the respective guide vane **62**. Each of these vane exterior surfaces **76** extends longitudinally along the vane mean line **70** between (and meets at) the respective vane leading edge **72** and the respective vane trailing edge **74**. Referring to FIG. **1**, each vane element **72**, **74**, **76**A, **76**B extends spanwise from the platform exterior surface **66** to the respective vane tip **68**.

(18) Each guide vane **62** may be configured as a fixed guide vane. Each guide vane **62**, for example, may be fixedly connected to the inner platform **60** and/or an internal support structure covered by the inner platform **60**. Alternatively, some or all of the guide vanes **62** may each be configured as a variable guide vane. Each guide vane **62**, for example, may be pivotally and/or otherwise moveably coupled to the inner platform **60** and/or the internal support structure covered by the inner platform **60** so as to be operable to change, for example, a pitch of the respective guide vane **62**.

(19) The guide vane structure **36** of FIG. **1** is disposed radially outboard of the turbine engine **30** and its engine housing **64**. This guide vane structure **36** is arranged axially aft and downstream of the forward end **58** of the turbine engine **30** and its engine housing **64**. The guide vane structure **36** of FIG. **1** thereby axially overlaps and circumscribes the turbine engine **30** and its engine housing **64**.

(20) The turbine engine **30** of FIG. **1** includes an inlet section **78**, a fan section **79**, a compressor section **80**, a combustor section **81**, a turbine section **82** and an exhaust section **83**. This turbine engine **30** also includes an (e.g., annular) inlet flowpath **86**, a (e.g., annular) core flowpath **88**, a (e.g., annular) bypass flowpath **90** and an exhaust flowpath **92**.

(21) The compressor section **80** of FIG. **1** includes a low pressure compressor (LPC) section **80**A and a high pressure compressor (HPC) section **80**B. The turbine section **82** of FIG. **1** includes a high pressure turbine (HPT) section **82**A, an intermediate pressure turbine (IPT) section **82**B and a power turbine (PT) section **82**C, which PT section **82**C is a low pressure turbine (LPT) section of the aircraft propulsion system **20** and its turbine engine **30**. At least (or only) the LPC section **80**A, the HPC section **80**B, the combustor section **81**, the HPT section **82**A, the IPT section **82**B and the PT section **82**C may collectively form a core **94** of the turbine engine **30**; e.g., a gas generator.

(22) The engine sections **78**, **79**, **80**A, **80**B, **81**, **82**A, **82**B, **82**C and **83** may be arranged sequentially along the axis **22** between the propulsion system forward end **24** and the propulsion system aft end **26**. With this arrangement, each engine section **80**A, **80**B of FIG. **1** is arranged axially along the axis **22** between (A) the propulsion module **28** and its members **34** and **36** and (B) the combustor section **81**. The fan section **79** of FIG. **1** is also arranged axially along the axis **22** between (A) the propulsor rotor **34** and (B) the engine core **94** and its engine sections **80**A-**82**C. The inlet section **78** of FIG. **1** is arranged at the engine forward end **58**. The exhaust section **83** of FIG. **1** is arranged at the propulsion system aft end **26**. The engine sections **79**-**82**C of FIG. **1** are arranged within the engine housing **64**, which may include an engine case **96** and a nacelle **98** over

the engine case **96**. The propulsor rotor **34** and the guide vane structure **36** are arranged outside of the engine housing **64**.

(23) Each of the engine sections **79**, **80**A, **80**B, **82**A, **82**B, **82**C includes a respective bladed rotor **100-105**; e.g., a ducted and/or shrouded engine rotor. Each of these bladed rotors **100-105** includes a rotor base (e.g., a disk or a hub) and a plurality of rotor blades (e.g., airfoils, vanes, etc.). The rotor blades are arranged circumferentially around the respective rotor base and the axis **22** in an array. The rotor blades may also be arranged into one or more stages longitudinally along the flowpath **86**, **88**. Each of the rotor blades is connected to the respective rotor base. Each of the rotor blades projects radially (e.g., spanwise) out from the respective rotor base into the flowpath **86**, **88** and to a distal tip of the respective rotor blade.

(24) The HPC rotor **102** is coupled to and rotatable with the HPT rotor **103**. The HPC rotor **102** of FIG. **1**, for example, is connected to the HPT rotor **103** by a high speed shaft **108**. At least (or only) the HPC rotor **102**, the HPT rotor **103** and the high speed shaft **108** collectively form a high speed rotating assembly **110**; e.g., a high speed spool of the turbine engine **30** and its engine core **94**. This high speed rotating assembly **110** is rotatable about the axis **22**. However, in other embodiments, the high speed rotating assembly **110** may be rotatable about another axis laterally and/or angularly offset from the axis **22**.

(25) The LPC rotor **101** is coupled to and rotatable with the IPT rotor **104**. The LPC rotor **101** of FIG. **1**, for example, is connected to the IPT rotor **104** by a low speed shaft **112**. At least (or only) the LPC rotor **101**, the IPT rotor **104** and the low speed shaft **112** collectively form a low speed rotating assembly **114**; e.g., a low speed spool of the turbine engine **30** and its engine core **94**. This low speed rotating assembly **114** is rotatable about the axis **22**. However, in other embodiments, the low speed rotating assembly **114** may be rotatable about another axis laterally and/or angularly offset from the axis **22**.

(26) The PT rotor **105** (e.g., the LPT rotor) is connected to and rotatable with a power turbine shaft **116**. At least (or only) the PT rotor **105** and the power turbine shaft **116** collectively form a power turbine rotating assembly **118**. This power turbine rotating assembly **118** is rotatable about the axis **22**. However, in other embodiments, the power turbine rotating assembly **118** may be rotatable about another axis laterally and/or angularly offset from the axis **22**.

(27) The fan rotor **100** is connected to and rotatable with a fan shaft **120**. At least (or only) the fan rotor **100** and the fan shaft **120** collectively form a fan rotating assembly **122**. This fan rotating assembly **122** is rotatable about the axis **22**. However, in other embodiments, the fan rotating assembly **122** may be rotatable about another axis laterally and/or angularly offset from the axis **22**.

(28) The fan rotating assembly **122** is coupled to the power turbine rotating assembly **118** through a first geartrain **124**; e.g., a transmission, a speed change device, an epicyclic geartrain, etc. This first geartrain **124** is disposed axially between and operatively couples the fan rotating assembly **122** and its fan shaft **120** to the power turbine rotating assembly **118** and its power turbine shaft **116**. With the foregoing arrangement, the fan rotor **100** is operable to rotate at a different (e.g., slower) rotational velocity than the power turbine rotating assembly **118** and its PT rotor **105**. A ratio of a gear system within the first geartrain **124** may be configured to increase efficiency of the fan section **79** as well as the PT section **82**C. The first geartrain **124** of FIG. **1** is located axially between (A) the engine core **94** and each of its engine sections **80**A-**82**C and (B) the propulsor rotor **34** and the fan rotor **100**.

(29) The propulsor rotor **34** is connected to and rotatable with a propulsor shaft **126**. At least (or only) the propulsor rotor **34** and the propulsor shaft **126** collectively form a propulsor rotating assembly **128**. This propulsor rotating assembly **128** is rotatable about the axis **22**. However, in other embodiments, the propulsor rotating assembly **128** may be rotatable about another axis laterally and/or angularly offset from the axis **22**.

(30) The propulsor rotating assembly **128** is coupled to the power turbine rotating assembly **118** through the first geartrain **124** and a second geartrain **130**; e.g., a transmission, a speed change

device, an epicyclic geartrain, etc. By contrast, the fan rotating assembly **122** of FIG. **1** is coupled to the power turbine rotating assembly **118** independent of the second geartrain **130**; e.g., not through/by way of the second geartrain **130**. The second geartrain **130** of FIG. **1**, for example, is disposed axially between and operatively couples the propulsor rotating assembly **128** and its propulsor shaft **126** to the fan rotating assembly **122** and its fan shaft **120** and, thus, to the power turbine rotating assembly **118** and its power turbine shaft **116** through the first geartrain **124**. Here, the fan shaft **120** is configured as an intermediate shaft/coupler between the first geartrain **124** and the second geartrain **130**. With the foregoing arrangement, the propulsor rotor **34** is operable to rotate at a different (e.g., slower) rotational velocity than (A) the fan rotating assembly **122** and its fan rotor **100** as well as (B) the power turbine rotating assembly **118** and its PT rotor **105**. A ratio of a gear system within the second geartrain **130** may be configured to increase efficiency of the propulsion module **28**. The second geartrain **130** of FIG. **1** is located axially between (A) the fan section **79** and its fan rotor **100** and (B) the propulsor rotor **34**. The fan section **79** of FIG. **1** and its fan rotor **100** are located axially between the first geartrain **124** and the second geartrain **130**.

(31) The inlet flowpath **86** fluidly couples an airflow inlet **132** into the turbine engine **30** from the external environment **38** to the core flowpath **88** and the bypass flowpath **90**. The inlet flowpath **86** of FIG. **1**, for example, extends longitudinally from the engine inlet **132**, through the fan section **79**, to an airflow inlet **134** into the core flowpath **88** and an airflow inlet **136** into the bypass flowpath **90**. The engine inlet **132** of FIG. **1** is located at the engine forward end **58** and/or axially next to and downstream of the propulsor rotor **34**.

(32) The core flowpath **88** extends longitudinally, in an axial aft direction away from the propulsor rotor **34**, through the engine core **94** from the inlet flowpath **86** to a combustion products core exhaust **138** from the core flowpath **88**. The core flowpath **88** of FIG. **1**, for example, extends sequentially through the LPC section **80**A, the HPC section **80**B, the combustor section **81**, the HPT section **82**A, the IPT section **82**B and the PT section **82**C from the core inlet **134** to the core exhaust **138**. The core inlet **134** of FIG. **1** is located downstream of and may be longitudinally/axially next to the fan section **79** and its fan rotor **100**. The core exhaust **138** of FIG. **1** is located downstream of and may be longitudinally/axially next to the PT section **82**C and its PT rotor **105**. The core inlet **134** and the core exhaust **138** may each be annular.

(33) The bypass flowpath **90** extends longitudinally, in the axial aft direction away from the propulsor rotor **34**, outside of the engine core **94** and its engine sections **80**A-**82**C from the bypass inlet **136** to a bypass exhaust **140** from the bypass flowpath **90**. The bypass flowpath **90** of FIG. **1**, for example, is disposed radially outboard of and extends axially along (e.g., axially overlaps) the first geartrain **124** and the engine core **94** from the bypass inlet **136** to the bypass exhaust **140**. The bypass inlet **136** of FIG. **1** is located downstream of and may be longitudinally/axially next to the fan section **79** and its fan rotor **100**. This bypass inlet **136** is disposed radially outboard of and may circumscribe the core inlet **134**. The bypass exhaust **140** may be located longitudinally/axially next to the PT section **82**C and its PT rotor **105**. This bypass exhaust **140** is disposed radially outboard of and may circumscribe the core exhaust **138**. The bypass inlet **136** and the bypass exhaust **140** may be annular. Alternatively, the bypass flowpath **90** may start off as annular and then may converge into a single non-annular leg or split into multiple non-annular legs outside of and extending along the engine core **94**. With such an arrangement, the bypass inlet **136** may be annular and each bypass exhaust **140** may be non-annular.

(34) The exhaust flowpath **92** of FIG. **1** fluidly couples the core flowpath **88** and the bypass flowpath **90** to an exhaust **142** from the turbine engine **30** into the external environment **38**. The exhaust flowpath **92** of FIG. **1**, for example, extends longitudinally from the core exhaust **138** and/or the bypass exhaust **140** to the engine exhaust **142**. The engine exhaust **142** of FIG. **1** is located at the propulsion system aft end **26**.

(35) With the foregoing arrangement, the core flowpath **88** and the bypass flowpath **90** are fluidly coupled in parallel between the inlet flowpath **86** and the exhaust flowpath **92**. However, in other

embodiments, it is contemplated the bypass flowpath **90** may alternatively be fluidly coupled to the external environment **38** independent of the exhaust flowpath **92**. The bypass exhaust **140**, for example, may alternatively be rerouted to adjoin the external environment **38**.

(36) During operation of the aircraft propulsion system **20**, ambient air within the external environment **38** is propelled by the propulsor rotor **34** across the guide vane structure **36** in an aft, downstream direction towards the propulsion system aft end **26**. A major outer portion of the air propelled by the propulsor rotor **34** flows across the guide vane structure **36** to provide forward thrust. The guide vane structure **36** and its guide vanes **62** condition (e.g., straighten out) the air propelled by the propulsor rotor **34**, for example, to remove or reduce circumferential swirl. A minor inner portion of the air propelled by the propulsor rotor **34** flows into the turbine engine **30** and its inlet flowpath **86** through the engine inlet **132**. The air within the inlet flowpath **86** is further propelled (e.g., compressed) by the fan rotor **100** and directed into the core flowpath **88** and the bypass flowpath **90**. The air entering the core flowpath **88** may be referred to as “core air”. The air entering the bypass flowpath **90** may be referred to as “bypass air”.

(37) The core air is compressed by the LPC rotor **101** and the HPC rotor **102** and directed into a combustion chamber **144** (e.g., an annular combustion chamber) of a combustor (e.g., an annular combustor) in the combustor section **81**. Fuel is injected into the combustion chamber **144** and mixed with the compressed core air to provide a fuel-air mixture. This fuel-air mixture is ignited and combustion products thereof flow through and sequentially drive rotation of the HPT rotor **103**, the IPT rotor **104** and the PT rotor **105** before being exhausted from the engine core **94** into the exhaust flowpath **92**. The rotation of the HPT rotor **103** and the IPT rotor **104** respectively drive rotation of the HPC rotor **102** and the LPC rotor **101** and, thus, compression of the air received from the core inlet **134**. The rotation of the PT rotor **105** drives rotation of the fan rotor **100** (through the first geartrain **124**) and rotation of the propulsor rotor **34** (through the first geartrain **124** and the second geartrain **130**). The rotation of the fan rotor **100** (A) propels the core air into the core flowpath **88** and (B) propels the bypass air through the bypass flowpath **90** into the exhaust flowpath **92**. The exhaust flowpath **92** subsequently directs the bypass air along with the combustion products out of the aircraft propulsion system **20** and its turbine engine **30** into the external environment **38** through the engine exhaust **142**. Concurrently, the rotation of the propulsor rotor **34** propels the ambient air within the external environment **38** across the guide vane structure **36** in the aft, downstream direction. With this arrangement, the turbine engine **30** and its engine core **94** power operation of (e.g., drive rotation of) the propulsor rotor **34** during aircraft propulsion system operation.

(38) The heat exchange system **32** of FIG. **1** includes one or more heat exchangers **146**. Each of these heat exchangers **146** may be arranged with (e.g., disposed in, project into, extend along, etc.) the bypass flowpath **90**. Each heat exchanger **146** is configured to transfer heat energy between the bypass air directed through the bypass flowpath **90** and a working fluid. Examples of the working fluid include, but are not limited to, the core air, the combustion products, lubricant, fuel, coolant, or the like. With the foregoing arrangement, the heat exchangers **146** utilize the bypass air rather than, for example, ambient air scooped from the external environment **38**, to enhance heat transfer efficiency. This may be particularly useful, for example, when the aircraft is operating on ground and air movement along the aircraft propulsion system **20** may be relatively slow and low pressure. By reducing or eliminating additional air scoop(s) along an exterior of the aircraft propulsion system **20**, free stream drag along the exterior of the aircraft propulsion system **20** may also be reduced. In addition, by directing the bypass air into the exhaust flowpath **92** following use with the heat exchange system **32**, remaining energy from the bypass air may be recuperated to supplement thrust generated by the combustion products exhausted through the engine exhaust **142** into the external environment **38**.

(39) In some embodiments, the bypass flowpath **90** may be configured with one or more sets of guide vanes (e.g., de-swirl vanes) disposed upstream and/or downstream of the heat exchangers

146.

(40) In some embodiments, one of the geartrains **124**, **130** may be a planetary geartrain and the other one of the geartrains **130**, **124** may be a star geartrain. The first geartrain **124** of FIG. **4**, for example, includes a star gear system. The second geartrain **130** of FIG. **4** includes a planetary gear system.

(41) Referring to FIG. **5**, the first geartrain **124** includes a first sun gear **148**, a first ring gear **150**, a plurality of first intermediate gears **152** (e.g., star gears) and a first carrier **154**. The first sun gear **148** is rotatable about the axis **22**. The first sun gear **148** is coupled to and rotatable with the power turbine rotating assembly **118** and its power turbine shaft **116**. The first ring gear **150** circumscribes the first sun gear **148** and the first intermediate gears **152**. The first ring gear **150** is rotatable about the axis **22**. The first ring gear **150** is coupled to and rotatable with the fan rotating assembly **122** and its fan shaft **120**; e.g., the intermediate shaft. The first intermediate gears **152** are arranged circumferentially about the axis **22** and the first sun gear **148** in an array. Each of the first intermediate gears **152** is disposed radially between and meshed with the first sun gear **148** and the first ring gear **150**. Each of the first intermediate gears **152** is rotatably mounted to the first carrier **154**. The first carrier **154** of FIG. **5** is fixedly connected to a stationary structure **156** of the aircraft propulsion system **20**.

(42) Referring to FIG. **6**, the second geartrain **130** includes a second sun gear **158**, a second ring gear **160**, a plurality of second intermediate gears **162** (e.g., planet gears) and a second carrier **164**. The second sun gear **158** is rotatable about the axis **22**. The second sun gear **158** is coupled to and rotatable with the fan rotating assembly **122** and its fan shaft **120**; e.g., the intermediate shaft. The second ring gear **160** circumscribes the second sun gear **158** and the second intermediate gears **162**. The second ring gear **160** of FIG. **6** is fixedly connected to a stationary structure **166** of the aircraft propulsion system **20**. The second intermediate gears **162** are arranged circumferentially about the axis **22** and the second sun gear **158** in an array. Each of the second intermediate gears **162** is disposed radially between and meshed with the second sun gear **158** and the second ring gear **160**. Each of the second intermediate gears **162** is rotatably mounted to the second carrier **164**. The second carrier **164** is rotatable about the axis **22**. The second carrier **164** is coupled to and rotatable with the propulsor rotating assembly **128** and its propulsor shaft **126**.

(43) Referring to FIG. **4**, the fan rotating assembly **122** and its fan shaft **120** may be rotatably supported by one or more bearings **168**; e.g., thrust bearings. Similarly, the propulsor rotating assembly **128** and its propulsor shaft **126** may be rotatably supported by one or more bearings **170**; e.g., thrust bearings.

(44) In some embodiments, referring to FIG. **7**, each of the geartrains **124** and **130** may be a planetary geartrain. The geartrains **124** and **130**, for example, may be configured as described above with respect to FIGS. **4-6**. However, whereas the first ring gear **150** of FIGS. **4-6** is rotatable and the first carrier **154** of FIGS. **4-6** is stationary, the first ring gear **150** of FIG. **7** is stationary and the first carrier **154** of FIG. **7** is rotatable. The first ring gear **150** of FIG. **7**, for example, is fixedly connected to the stationary structure **156** of the aircraft propulsion system **20**. The first carrier **154** is rotatable about the axis **22**. Moreover, the first carrier **154** is coupled to and rotatable with the fan rotating assembly **122** and its fan shaft **120**; e.g., the intermediate shaft.

(45) In some embodiments, referring to FIG. **8**, each of the geartrains **124** and **130** may be a star geartrain. The geartrains **124** and **130**, for example, may be configured as described above with respect to FIGS. **4-6**. However, whereas the second ring gear **160** of FIGS. **4-6** is stationary and the second carrier **164** of FIGS. **4-6** is rotatable, the second ring gear **160** of FIG. **8** is rotatable and the second carrier **164** of FIG. **8** is stationary. The second carrier **164** of FIG. **8**, for example, is fixedly connected to the stationary structure **166** of the aircraft propulsion system **20**. The second ring gear **160** of FIG. **8** is rotatable about the axis **22**. Moreover, the second ring gear **160** is coupled to and rotatable with the propulsor rotating assembly **128** and its propulsor shaft **126**.

(46) The aircraft propulsion system **20** may have various propulsion module configurations other

than the one described above. For example, referring to FIG. **9**, the propulsion module **28** may be configured with the single propulsor rotor **34** and without a guide vane structure. In another example, referring to FIG. **10**, the propulsion module **28** may be configured with multiple of the propulsor rotors **34**A and **34**B (generally referred to as "**34**") (e.g., counter-rotating open propulsor rotors). Each of these propulsor rotors **34** may be rotationally driven by the turbine engine **30** and its power turbine rotating assembly **118** through the geartrains **124** and **130** (see FIG. **1**). In the dual propulsor rotor arrangement of FIG. **10**, the propulsion module **28** is configured without a guide vane structure. The present disclosure therefore is not limited to the foregoing exemplary propulsor module configurations.
(47) The aircraft propulsion system **20** of FIG. **1** is described above with a tractor configuration; e.g., where each propulsor rotor **34** is disposed at or otherwise near the propulsion system forward end **24**. It is contemplated, however, the aircraft propulsion system **20** may be reversed to provide a pusher fan configuration. The present disclosure therefore is not limited to any particular open rotor propulsion system arrangement.
(48) The aircraft propulsion system **20** of FIG. **1** is described above with three rotating assemblies within the turbine engine **30** and its engine core **94**. It is contemplated, however, the turbine engine **30** and its engine core **94** may alternatively include more than three rotating assemblies or less than three rotating assemblies. For example, the turbine engine **30** and its engine core **94** may omit the PT section **82**C and its associated rotor **105**. In such embodiments, the turbine section **82**B may be configured as a low pressure turbine section of the turbine engine **30** and its engine core **94**. Here, the turbine rotor **104** may replace the turbine rotor **105**, and the shaft **112** may be coupled to the first geartrain **124** rather than the shaft **116**. Of course, various other arrangements are possible and contemplated.
(49) While various embodiments of the present disclosure have been described, it will be apparent to those of ordinary skill in the art that many more embodiments and implementations are possible within the scope of the disclosure. For example, the present disclosure as described herein includes several aspects and embodiments that include particular features. Although these features may be described individually, it is within the scope of the present disclosure that some or all of these features may be combined with any one of the aspects and remain within the scope of the disclosure. Accordingly, the present disclosure is not to be restricted except in light of the attached claims and their equivalents.

Claims

1. A propulsion system for an aircraft, comprising: an open propulsor rotor; and a turbine engine including a fan section, an engine core, a first geartrain, a second geartrain, a core flowpath and a bypass flowpath; the fan section comprising a fan rotor; the engine core including a first rotating assembly, a low pressure compressor section, a high pressure compressor section, a combustor section, a high pressure turbine section and a low pressure turbine section, the low pressure compressor section is located axially between the first geartrain and the high pressure compressor section along the axis, and the first rotating assembly rotatable about an axis and configured to drive rotation of the open propulsor rotor and the fan rotor through the first geartrain; the first geartrain located axially between the engine core and the open propulsor rotor, the first rotating assembly configured to drive the rotation of the fan rotor through the first geartrain and independent of the second geartrain; the second geartrain located axially between the fan rotor and the open propulsor rotor; the core flowpath extending through the low pressure compressor section, the high pressure compressor section, the combustor section, the high pressure turbine section and the low pressure turbine section from an inlet into the core flowpath to an exhaust from the core flowpath, and the inlet into the core flowpath next to and downstream of the fan section; and the bypass flowpath extending outside of and along the engine core from an inlet into the bypass

flowpath to an exhaust from the bypass flowpath, the exhaust from the bypass flowpath disposed radially outboard of and circumscribing the exhaust from the core flowpath, and the inlet into the bypass flowpath next to and downstream of the fan section.

2. The propulsion system of claim 1, wherein the first rotating assembly comprises a low pressure turbine rotor in the low pressure turbine section.

3. The propulsion system of claim 1, wherein the engine core further includes an intermediate pressure turbine section between the high pressure turbine section and the low pressure turbine section along the core flowpath.

4. The propulsion system of claim 3, wherein the intermediate pressure turbine section is configured to rotate independent of the first rotating assembly.

5. The propulsion system of claim 1, wherein the low pressure turbine section comprises a low pressure turbine rotor; and the first geartrain is located axially between the low pressure turbine rotor and the fan rotor.

6. The propulsion system of claim 1, wherein the first rotating assembly is configured to drive the rotation of the open propulsor rotor through the first geartrain and the second geartrain.

7. The propulsion system of claim 6, wherein one of the first geartrain and the second geartrain comprises a planetary gear system; and the other one of the first geartrain and the second geartrain comprises a star gear system.

8. The propulsion system of claim 6, wherein the first geartrain comprises a planetary gear system; and the second geartrain comprises a planetary gear system.

9. The propulsion system of claim 6, wherein the first geartrain comprises a star gear system; and the second geartrain comprises a star gear system.

10. The propulsion system of claim 6, wherein the fan rotor is located axially between the first geartrain and the second geartrain.

11. The propulsion system of claim 1, wherein at least a portion of the bypass flowpath is annular.

12. The propulsion system of claim 1, further comprising a heat exchanger disposed in the bypass flowpath.

13. The propulsion system of claim 1, wherein the turbine engine further includes an exhaust flowpath fluidly coupling the core flowpath and the bypass flowpath to an exhaust from the turbine engine.

14. The propulsion system of claim 1, further comprising an open guide vane structure downstream of the open propulsor rotor, the open guide vane structure configured to condition air propelled by the open propulsor rotor.

15. The propulsion system of claim 1, further comprising: a second open propulsor rotor; the first rotating assembly further configured to drive rotation of the second open propulsor rotor through the first geartrain.

16. A propulsion system for an aircraft, comprising: a first gear system including a first sun gear, a first ring gear, a plurality of first intermediate gears and a first carrier, each of the plurality of first intermediate gears meshed with and radially between the first sun gear and the first ring gear, and each of the plurality of first intermediate gears rotatably mounted to the first carrier; a second gear system including a second sun gear, a second ring gear, a plurality of second intermediate gears and a second carrier, each of the plurality of second intermediate gears meshed with and radially between the second sun gear and the second ring gear, each of the plurality of second intermediate gears rotatably mounted to the second carrier; an intermediate shaft coupled to the first ring gear and the second sun gear; a fan rotor coupled to the intermediate shaft, the fan rotor located axially between the first gear system and the second gear system; an open propulsor rotor coupled to the second carrier; and an engine core comprising a rotating assembly of a turbine engine, the rotating assembly coupled to the first sun gear, the rotating assembly configured to drive rotation of the fan rotor through the first gear system and independent of the second gear system, and the rotating assembly configured to drive rotation of the open propulsor rotor through the first gear system and

the second gear system; wherein the first gear system is arranged between the engine core and the fan rotor.

17. A propulsion system for an aircraft, comprising: a plurality of gear systems, each of the plurality of gear systems comprising a planetary gear system or a star gear system; an intermediate shaft operatively coupling a first of the plurality of gear systems to a second of the plurality of gear systems; a fan rotor coupled to the intermediate shaft; an open propulsor rotor operatively coupled to the second of the plurality of gear systems; and an engine core comprising a rotating assembly of a turbine engine, the rotating assembly operatively coupled to the first of the plurality of gear systems, the rotating assembly configured to drive rotation of the fan rotor through the first of the plurality of gear systems and independent of the second of the plurality of gear systems, and the rotating assembly configured to drive rotation of the open propulsor rotor through the first of the plurality of gear systems and the second of the plurality of gear systems; wherein the first of the plurality of gear system is located axially between the engine core and the fan rotor.